

Line-Level Manuscript Review

Two Companion Papers on Quantum Channel and Instrument Compression Bounds

Manuscript A: main-article-revised2.txt (3,646 lines, ~16,600 words)

Manuscript B: instruments-paper-revised2.txt (1,615 lines, ~13,900 words)

Source: repository channel-supp-augmented, folder manuscript uploads

Review date: 8 September 2026

Method: two full reading passes, independent re-derivations, automated checks

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1. Review Scope, Materials, and Method

This report presents a line-level review of the two LaTeX manuscripts stored in the folder manuscript uploads of the repository channel-supp-augmented, retrieved on 8 September 2026: **main-article-revised2.txt**, the channel paper “Exact Diamond Balls, Antipodal Widths, and Query-Uniform Compression Bounds for Quantum Channels” (3,646 lines, approximately 16,600 words), and **instruments-paper-revised2.txt**, the instruments paper “Exact Affine Geometry, Optimal Centres, and Certified Compression Bounds for Finite-Outcome Quantum Instruments” (1,615 lines, approximately 13,900 words). All line numbers in this report refer to the uploaded .txt sources exactly as retrieved; no line numbering was introduced or altered. Where this report refers to figures and tables by name (for example fig:replacement-join or tab:summary), the LaTeX label is used so that the reference is unambiguous regardless of float placement.

The mandate was to read and re-read both articles thoroughly at line level for flaws, internal inconsistencies, and opportunities for improvement, and to provide a report without modifying them. The originals were not modified: all analysis was performed on local copies fetched through the repository's read interface, and nothing was written back. One housekeeping note: the repository access credential was shared in plaintext chat in order to retrieve the files; it served its purpose and should now be rotated or revoked as standard hygiene.

The review proceeded in three passes. **Pass 1** was a full structural read of both papers from first line to last, registering every theorem, lemma, proposition, corollary, definition, remark, figure, table, and worked example together with every quantitative claim they contain. **Pass 2** was a line-level re-read in which every recomputable constant was recomputed independently (worked examples, envelope breakpoints, table entries, figure annotations, asymptotic limits) and the proofs of the central statements were re-derived; the resulting verification log is recorded in Section 4. **Pass 3** applied automated tooling: both sources were compiled end to end with a standard LaTeX engine (Tectonic), and static integrity checks were run for cross-reference resolution, citation-key matching, environment and brace balance, non-ASCII characters, and spelling-convention consistency. Sections 3 through 9 report the findings; Section 10 gives the overall assessment.

2. Executive Summary

The mathematical content of both manuscripts survives a hostile line-level read. Every numerical claim that was independently recomputed - all envelope breakpoints, both worked examples, all table entries, the asymptotic product limit, and the complementarity identities -

matched exactly, and the re-derivations of the sampled proofs found no mathematical errors. Both papers compile cleanly with a standard engine, with no undefined references, no missing citations, and no structural defects. The defects that were found are concentrated in presentation and consistency rather than correctness, and none of them threatens a theorem.

The strengths are worth stating explicitly, because they shape the recommendations. Both papers maintain a disciplined separation between exact results and certified bounds (the exact-versus-certified remark in the instruments paper, lines 157-160, and the corresponding scoping remarks in the channel paper); both handle degenerate cases carefully (the $d_B = 1$ singleton channel body, the $n = d_B = 1$ degenerate instrument, empty latent spaces, the $+\infty$ coindex convention, and the diameter cap on the finite-sample bound); and the cross-referencing between the two companion papers is accurate in both directions. This scaffolding materially reduces referee friction and should be preserved through revision.

The defects cluster in four areas. First, one figure in the channel paper (fig:replacement-join, lines 1104-1147) imports the flag-space notation C_n and the outcome count n from the companion instruments paper, neither of which is defined anywhere in the channel paper. Second, the channel paper overloads the symbols s , q , and Δ across unrelated meanings and mixes British and American spellings on more than twenty-five lines, including the same technical term spelled both ways. Third, a small set of external citations could not be verified offline and should be checked against the published records; one of them is load-bearing for a lemma, although this review re-derived that lemma's inequality independently and confirmed it. Fourth, the two companion papers are asymmetric in several presentational respects (Acknowledgements, hyperlink style, definition numbering, and two mathematical statements present in only one paper) that an editor will notice when handling them as concurrent submissions. The severity-ranked table below summarizes all findings; full details with line references follow in Sections 5 and 6.

Table 1. Summary of findings by severity

ID	Severity	Location	Finding in brief
A1	High	A, lines 1104-1147	Figure 2 (fig:replacement-join) uses the flag space C_n and outcome count n , which are undefined in the channel paper.
A2	High	A, lines 782 vs 3380, 1143	Symbol s means $\min\{d_A, d_B\}$ in the radius theorems but block size in the partition appendix and a figure caption.

ID	Severity	Location	Finding in brief
A3	High	A, lines 1161 vs 3380; 472, 746, 2585, 2887	Symbols q and Δ each carry two to four unrelated meanings (log-index vs block count; diameter vs perturbation vs simplex).
A4	High	A, 25+ lines	Mixed British and American spelling, including “normalized Choi operator” (line 628) vs “normalised Choi operators” (line 1291).
A5	Medium	A, lines 712-734	Load-bearing upper bound in Lemma 3.2 rests on an external citation; a three-line self-contained proof exists (see Section 4.3).
A6	Medium	A, line 50	Abstract says “Matching upper bounds ... bracket the intermediate widths”; the bounds bracket but do not match.
A7	Medium	A, lines 1762-1764	Duplicated decoder-definition sentence with a dangling “as above” in the proof of the output-pinching theorem.
A8	Medium	A, lines 1159-1169	Compound Clifford theorem merges two constructions and rebinds the parameter p mid-statement.
A9	Medium	A, lines 1099-1102 and App. partition section	Structured-partition appendix is strictly weaker than the replacement-sphere theorem and is acknowledged as such.

ID	Severity	Location	Finding in brief
A10	Medium	Both papers	Companion asymmetries: Acknowledgements, hyperref style, definition numbering, missing channel-paper analogues of two instruments-paper statements.
B1	Medium	B, line 928	“Kolmogorov n-widths” collides with the outcome count n; “continuous images of R” is imprecise.
B2	Medium	Both papers	External citation records to verify before submission (Section 8, Table 4).
A11 / B6	Low	A and B, compile log	Overfull lines and systematic underfull boxes in the p-column tables of both papers.
A12	Low	A, lines 3523-3643	Bibliography neither in citation order nor alphabetical; 24 labels never cross-referenced.

3. Compilation and Automated Integrity Checks

Both sources were compiled end to end with Tectonic (a standard, self-contained LaTeX engine using the TeX Live bundle), starting from the uploaded files renamed to .tex with no other edits. Both compiled to completion with exit code 0 and produced PDFs (322 KB and 302 KB respectively), with no errors, no undefined references, and no missing citations in either log. This is a meaningful baseline: many pre-submission manuscripts fail at this stage, and neither of these does.

The compile logs contain only cosmetic warnings. The channel paper shows four overfull horizontal boxes (source lines 1528-1531, 2065-2068, 2388-2391, and 2771-2775, the worst being 14.4 points at 2065-2068) and systematic underfull boxes inside its two p-column tables (tab:channel-scales at lines 1939-1983 and tab:resource-models at lines 2551-2593). The instruments paper shows four overfull boxes (lines 49-50, 712-714, 747-748, and 1557-1558, the worst being 11.6 points at 712-714) and the same pattern of underfull boxes in its wide tables (tab:summary at 1281-1298 and tab:comparison at 1437-1456). Underfull boxes

in p-columns are the familiar cost of justified narrow columns; the standard fix (ragged-right columns, or restructuring to natural-width columns) is listed as a low-priority recommendation.

Static integrity checks were equally clean. Every cross-reference resolves: the channel paper defines 96 labels and uses 72 of them, the instruments paper 77 and 57, with no dangling references in either direction. Every citation key has a matching bibliography entry and every bibliography entry is cited (23 of 23 in the channel paper, 16 of 16 in the instruments paper; one channel-paper citation uses an optional argument, cite with “Corollary 2”, and resolves correctly). Environments are balanced, braces balance, no display-math dollar-pairs are used, and both files contain zero non-ASCII bytes, with all diacritics written as proper LaTeX escape sequences. One robustness note: the channel paper uses `resizebox` (line 2622) without explicitly loading `graphicx`; the compile succeeds because the loaded TikZ/PGF chain supplies the command, but a directly used command should have its package loaded explicitly, since package internals are not a stable contract across distributions.

Table 2. Automated check results

Check	Manuscript A (channels)	Manuscript B (instruments)
Full compile (Tectonic)	Exit 0; 322 KB PDF; no errors	Exit 0; 302 KB PDF; no errors
Undefined references / citations	None	None
Overfull hboxes (source lines)	4: 1528-1531, 2065-2068, 2388-2391, 2771-2775	4: 49-50, 712-714, 747-748, 1557-1558
Underfull hboxes	Tables 1-2 (lines 1939-1983, 2551-2593)	Tables 1-2 (lines 1281-1298, 1437-1456)
Labels / references used	96 / 72, all resolve	77 / 57, all resolve
Citation keys vs bibitems	23 / 23, all resolve	16 / 16, all resolve
Environments, braces, dollar-pairs	Balanced; none	Balanced; none
Non-ASCII bytes	0 (diacritics via escapes)	0 (diacritics via escapes)

4. Mathematical Verification Log

This section records what was independently verified, so that the findings in Sections 5 and 6 can be read with a known baseline of confidence. Two kinds of verification were performed: recomputation of every constant that appears in worked examples, tables, and figures; and re-derivation of the proofs of the central theorems. No discrepancy was found in either category. All notation below follows the papers' own conventions.

4.1 Constants, examples, tables, and figures

Every numeric claim in the two worked examples, the two summary tables of the instruments paper, and the four envelope figures was recomputed from the papers' own formulas. The results are collected in Table 3. In addition to the tabulated items, the algebraic side conditions were checked: the join extends the value-one range exactly when d_A^2 is at least $4d_B - 2$ (so the claim “for every square system of dimension at least four” is correct); the two-term trade-off of the intermediate floor crosses at n of order $2d_A / (d_B \min\{d_A, d_B\})$; and for $d_A \leq d_B$ with $d_B \geq 3$ the term $1/d_A$ strictly dominates the centred-ball term, as the text asserts.

Table 3. Recomputed quantities (all confirmed)

Paper	Item (source lines)	Recomputed values	Status
A	Qubit worked example (1987-2012)	$D = 12$; constant-code width $3/2$; $L = 2$; floor $1/2$; balanced pinching threshold $r = 4$	Confirmed
A	Qubit envelope figure (2014-2050)	Segments 1, 0.5, 1.5, 1; convex-projection value 1.8 at $r = 3$; drop at $r = 4$	Confirmed
A	Chebyshev radius and identity (1600-1630)	$R = 2(1 - 1/(d_B s))$; in-radius + covering radius = 2	Confirmed
B	Worked example (2,2,2) (1336-1348)	$D = 28$; $L_{\text{rep}} = 6$, $L_{\text{obs}} = 2$, $N_{\text{join}} = 2$, $L_{\text{join}} = 3$, $L = 6$; floor $1/2$; $R_n = 1.75$; thresholds 12 and 14	Confirmed
B	Envelope figure panel (b), (3,2,2) (1381-1408)	$D = 63$; $L = 8$ from the join; floor $1/3$; input pinching at 21 (value $4/3$), output pinching at 27 (value 1)	Confirmed
B	Threshold table rows (2,3,1) and (2,1,2) (1458-1474)	$D = 32$, $L = 7$, floor $1/2$; $D = 4$, $L = 2$, floor 1; collapse case $n d_B s = 2$	Confirmed
B	Asymptotic product (1237-1257)	Limit $2d_A \max\{d_A, d_B\}$, approached from below	Confirmed
A+B	All $n = 1$ specializations	Dimension, in-radius, covering radius, sphere indices, and floor $\max\{\rho, 1/d_A\}$ all agree across the two papers	Confirmed

4.2 Proof-level re-derivations

The following proofs were re-derived in full during the second pass and found correct. In the channel paper: the exact fiber-radius identity and its factor-two comparison; the antipodal-profile lower bound via the Borsuk-Ulam collision argument; the deleted-product coindex bounds, including the normalized-difference equivariant map and the convex-body equality case; the Helly-type convex-projection bound, including the barycenter estimate that $(m-1)/m \leq k/(k+1)$ for $m \leq k+1$; the stable source-latent obstruction inequality; the radial construction for contained balls; the exact in-radius theorem's two halves (the spectral bound $\lambda_{\max}(-X) \leq (s / 2 d_A)$ times the diamond norm, together with the restricted maximally entangled input identity, and the isometry witness with traceless unitary conjugation); the Chebyshev domination argument (the rank bound on bipartite states, the $a \leq K b$ trace argument, and the pure isometric output witness); the pinching identity through the roots-of-unity average and the spectrum of the all-ones matrix; the balanced-partition exchange algebra (the three displayed changes $2ab + 1$, $2(a - 1)(b - 1)$, and $2(s - a)(s - b)$, all verified by expansion); the hybrid Lipschitz estimate with the stabilization argument at reference dimension d_A ; the postselection normalization constant $2N / p_0$; the packing volume bound $\eta \leq 4 / (L^{1/n} - 1)$; the qubit Bloch-sphere tetrahedron bound; the sampled-transcript lower bound via Borsuk-Ulam on the simplex; the informationally complete POVM positivity margin; the finite-sample estimator chain, including the multinomial covariance identity; the Clifford anticommutation relations and the orthogonal-support property of the antipodal states; and the Weyl-unitary orthogonality.

In the instruments paper, the corresponding statements were re-derived with the same outcome: the affine-dimension count with the surjective marginal map; the instrument spectral bound and its two-witness maximality argument in both regimes (isometry witness for $d_B > 1$, effect-transfer witness for $d_B = 1$); the covering-radius domination with the group action of the output unitaries together with outcome permutations; the concavity and Lipschitz properties of the in-radius; the replacement-instrument isometry and the exact replacement profile; the full-direction sphere with the correction state; the observable sphere in both the $n \geq 2$ flagged form and the $n = 1$ channel form; the instrument-level join, including the single-input exposure argument for why exactly one input-dependent sphere can enter; the collapse theorem in both degenerate cases; the zero-error threshold with the continuous nearest-point projection decoder; and the no-programming appendix arguments (mixed-program orthogonality, unequal-output reduction, and the compactness divergence argument with its subsequence discipline).

4.3 The Lemma 3.2 upper bound, re-derived

One step of the channel paper was flagged for closer inspection because it is both load-bearing and externally cited: the upper bound in Lemma 3.2 (lines 712-734), which states

that for a Hermiticity-preserving map Δ with unnormalized Choi operator J_Δ , the diamond norm is at most the operator norm of the partial trace of the absolute value, written $\|\Delta\| \leq \|\text{Tr}_B |J_\Delta| \|_\infty$. The bound is used in the proof of the identity-versus-depolarizing distance (supplying the matching upper bound $2(1 - 1/d^2)$) and as the first inequality of the finite-sample estimator. This review re-derived the inequality and confirms it; the argument is short enough to include verbatim as a recommendation.

Write the Jordan decomposition of the unnormalized Choi operator as $J = J_+ - J_-$ with both parts positive, and let Δ_+ and Δ_- be the corresponding completely positive maps. For every state ρ on a reference-and-input system, the image $(\text{id} \otimes \Delta)(\rho) = (\text{id} \otimes \Delta_+)(\rho) - (\text{id} \otimes \Delta_-)(\rho)$ is a difference of two positive operators, so its trace norm is at most the sum of the traces. Computing each trace through the adjoint gives $\text{Tr}[(\text{id} \otimes \Delta_\pm)(\rho)] = \text{Tr}[\rho (\text{I} \otimes (\text{Tr}_B J_\pm)^T)]$, so the sum equals $\text{Tr}[\rho (\text{I} \otimes (\text{Tr}_B |J|)^T)]$, which is at most the operator norm of $\text{Tr}_B |J|$ because ρ is a state. Taking the supremum over ρ yields exactly the stated bound. The recommendation that follows (finding A5) is to include this three-line argument in the paper and to verify the external attribution.

5. Findings - Manuscript A (Channel Paper)

Manuscript A is the channel paper, “Exact Diamond Balls, Antipodal Widths, and Query-Uniform Compression Bounds for Quantum Channels.” The findings below are ordered by severity; each records the location, the observation, and a concrete recommendation. Findings marked High are internal inconsistencies visible to a careful first read; Medium findings are wording, structure, or verification matters with referee impact; Low findings are editorial polish.

A1 - Figure 2 imports companion notation that is undefined in this paper

Severity: High. Location: Channel paper, lines 1104-1147 (Figure 2, fig:replacement-join); single occurrence of C_n at line 1119.

The TikZ node at line 1119 reads “ $B \otimes C_n$: flag-diagonal $n d_B^2 \dim$ ”. The classical flag space C_n and the outcome count n are defined only in the companion instruments paper; a search of the channel source confirms that C_n appears exactly once, in this figure, and is never defined. The caption (line 1145) legitimately cross-references the n -outcome analogue in the companion paper, but the body of the figure itself cannot be interpreted from the channel paper alone. This is an internal inconsistency of the kind that porting material between companion drafts produces, and it is the single clearest defect found in either manuscript.

Recommendation. Redraw the node in channel-only terms, for example as the block decomposition of the output space with the real dimension count of its Hermitian operators,

or add one sentence in Section 5.1 defining the flag convention used in the figure. Either fix takes minutes and removes the only undefined symbol in the paper.

A2 - Symbol s carries two incompatible meanings

Severity: High. Location: Channel paper, line 782 (exact in-radius theorem) and passim, versus lines 3380 and 3411, and the Figure 2 caption at line 1143.

Throughout the radius and covering results, s denotes $\min\{d_A, d_B\}$ and appears in every headline formula (the in-radius $2/(d_B s)$, the covering radius $2(1 - 1/(d_B s))$). In the structured-partition appendix, the same letter denotes the minimum block size (“Fix $2 \leq s \leq d_B$ ”, lines 3380 and 3411), and the Figure 2 caption uses it in that second sense (“ $s=2$ gives $\alpha_r=1$ for $0 \leq r \leq 3 \text{ floor}(d_B/2) - 1$ ”). Both uses are locally correct, but the two live in the same paper one section apart, and a reader who has memorized $s = \min\{d_A, d_B\}$ from Section 4 will misread the appendix and the caption.

Recommendation. Rename the block-size parameter (to b or to a symbol such as σ_{blk}), and add a short notation remark. The companion instruments paper already contains exactly such a remark (its notation remark, lines 162-165, which explicitly de-collides ranks, block counts, latent dimensions, and covering radii); mirroring it in the channel paper would also improve symmetry between the companions.

A3 - Symbols q and Delta are each overloaded

Severity: High. Location: Channel paper: q at lines 1161 and 1176 versus lines 3380 and 3450; Delta at lines 472-474, throughout Sections 3-5, line 2585, and line 2887.

The letter q denotes $\text{floor}(\log_2 d_B)$ in the Clifford theorem and its corollary (lines 1161 and 1176) and, unrelatedly, the number of full blocks in the partition corollary (“ $d_B = q s + t$ ”, lines 3380 and 3450). The letter Delta is used with four distinct meanings: the diameter of the source in the convex-projection proposition (lines 472-474); Hermiticity-preserving perturbation maps throughout Sections 3 to 5; the transcript simplex in the resource-models table (line 2585); and the probability simplex in the one-query encoding proposition (line 2887, which even introduces a compound notation Delta_D for it). No individual use is erroneous, but four meanings of one glyph across a 3,600-line paper is an error factory, particularly because the perturbation and simplex meanings of Delta meet inside the same subsection on sampled transcripts.

Recommendation. Use distinct symbols for the simplex (a plain capital Sigma with a subscript, or the paper's own $\Delta(c)$ style with a definition), keep the diameter on the existing diam macro, and disambiguate the two uses of q . A one-page symbol table in an appendix would resolve all residual ambiguity and is inexpensive.

A4 - Mixed British and American spelling conventions

Severity: High. Location: Channel paper, 25+ lines; representative pairs at lines 50 vs 98, 628 vs 1291, and 1270 vs 628.

An automated scan with manual confirmation shows the channel paper uses “centred” on ten lines (50, 1338, 1345, 1379, 1433, 1468, 1487, 1960, 1961, 3027) and “centered” or “Center” on fifteen (98, 115, 120, 498, 737, 776, 778, 828, 830, 861, 864, 959, 980, 1930, 3054, and others), and “normalized” on fourteen lines against “normalised” at line 1291 and “Normalisation” at line 1270. The clearest exhibit is that the same technical term appears both ways: “normalized Choi operator” at line 628 versus “normalised Choi operators” at line 1291. The companion instruments paper is uniformly British (“centred”, “normalised”, “minimises”, “specialises”, and “analysed” in its Data Availability statement), which makes the pair look inconsistent as co-submissions.

Recommendation. Unify the channel paper internally, and align the two papers on one convention. British spelling requires fewer edits in the instruments paper and American fewer in the channel paper; the choice matters less than consistency, but since the papers cite each other and will be handled together, both should match.

A5 - Load-bearing external citation in Lemma 3.2

Severity: Medium. Location: Channel paper, lines 712-734 (Lemma 3.2); consequences at lines 927-941 and 2965-2973.

The upper bound of Lemma 3.2 is attributed to a corollary of the Nechita-Puchala-Pawela-Zyczkowski paper, with an explicit note about the unnormalized Choi convention. Two concerns arise. First, this review could not verify the published statement offline, and the citation is load-bearing: the bound supplies the matching upper bound in the identity-versus-depolarizing corollary and the first inequality of the finite-sample estimator, so any mismatch in convention or statement would propagate. Second, the inequality admits a three-line self-contained proof, recorded in Section 4.3 of this report, which this review verified.

Recommendation. Include the self-contained proof in the paper and retain the citation as a “see also”. This removes the load-bearing dependency on an external statement and any convention-ambiguity risk, at the cost of four lines of text.

A6 - Abstract overstates the upper bounds

Severity: Medium. Location: Channel paper, line 50.

The abstract states: “Matching upper bounds from convex projection and balanced pinching bracket the intermediate widths.” The upper bounds bracket the lower envelope but do not match it; the gap is the paper’s own declared open problem, stated in the certified-envelope remark, the interval corollary, the exact-versus-non-exact remark, and item 1 of the

conclusion's open problems. A referee reading “matching” in the abstract and then encountering an explicitly open gap in the body may read the abstract as overclaiming.

Recommendation. Replace “Matching” with “Complementary constructive” or simply “Upper bounds ... bracket the intermediate widths”.

A7 - Duplicated decoder sentence with a dangling reference

Severity: Medium. Location: Channel paper, lines 1762-1764, proof of the input/output pinching theorem.

The proof of the output-pinching bound contains two consecutive sentences that define the same decoder twice: “Invert the coordinates on the encoded image and use any fixed channel, e.g., Ω , elsewhere. Define the decoder as above on the encoder image and set it equal to that fixed channel outside.” The second sentence restates the first and refers to “as above” although no decoder has been defined above this point in the proof. This is clearly an editing remnant.

Recommendation. Delete the second sentence (or the first, keeping the more explicit one). One-minute fix.

A8 - Compound statement in the Clifford theorem

Severity: Medium. Location: Channel paper, lines 1159-1169 (Clifford-algebra theorem).

The theorem merges two constructions in one statement: a replacement-channel family on a sphere for arbitrary input-output dimensions, and a unitary-channel family for square systems. The parameter p is bound twice with different values (“Let $q = \text{floor}(\log_2 d_B)$, $p = 2^q$ ” and then “If $A = B = H$ is square of arbitrary dimension $d \geq 2$, choose $p = 2^{\text{floor}(\log_{\{2\}} d)}$ ”), which is technically consistent but reads as two theorems fused together. The proof in the appendix is clean and covers both parts, so this is purely a statement-structure issue.

Recommendation. Split into a proposition (replacement sections, general dimensions) and a theorem (unitary sections, square systems), or restructure so that the parameters are bound once and the square-system clause is a corollary.

A9 - Structured-partition appendix is redundant relative to the replacement sphere

Severity: Medium. Location: Channel paper, lines 1099-1102, and the partition section of the appendix (lines 3314-3461).

The main text concedes the point explicitly: the structured partition constructions are “retained as explicit low-rank subfamilies, although [the replacement-sphere theorem] provides the stronger unrestricted replacement-channel bound.” The appendix then devotes roughly 150 lines to a join construction, a partition theorem, a block-size envelope, and an

optimization proposition whose headline index $R = \Sigma(b_i^2 - 1) - 1$ never exceeds $d_B^2 - 2$, the replacement theorem's index. The material is mathematically correct (the exchange-algebra computations were verified), but as presented a referee may ask why approximately five results are included that are uniformly dominated by a theorem already proved in the main text.

Recommendation. Either demote the appendix to a compact remark presenting the block join and the balanced-block optimization in a few lines, or add one forward-looking sentence justifying the retention (the block structure is the channel-level shadow of the flag-block partitions that govern the instrument-level theory in the companion paper, and the optimization is reused there). As written, the justification “explicit low-rank subfamilies” is thin.

A10 - Presentational asymmetries between the companion papers

Severity: Medium. Location: Both papers; details below.

Six asymmetries were identified that an editor handling the two papers as concurrent co-submissions will notice. (i) The instruments paper contains an Acknowledgements section withheld for review (lines 1485-1486); the channel paper has none. (ii) Hyperlink configuration differs: colorlinks in the channel paper (line 15) versus hidelinks in the instruments paper (line 10), producing visibly different PDFs. (iii) Definition numbering differs: the channel paper numbers definitions on a separate counter from theorems, while the instruments paper shares one counter, so cross-paper references by number will not align in style. (iv) The instruments paper states an explicit error-one-floor proposition (every certified upper-bound term is at least one below the affine dimension); the channel paper contains the same mathematical content implicitly but never states it, although a one-line corollary would do so. (v) The instruments zero-error theorem strengthens the threshold with a continuous decoder at and above the affine dimension; the channel paper's threshold proposition lacks this refinement even though the identical nearest-point-projection argument applies verbatim. (vi) The Data Availability statements differ in spelling convention (“analyzed” versus “analysed”), which combines with finding A4.

Recommendation. Harmonize all six items before submission. Items (iv) and (v) are genuine mathematical improvements to the channel paper at negligible cost; the rest are cosmetic but cheap.

A11 - Typesetting warnings in the channel paper

Severity: Low. Location: Channel paper, compile log; overfull lines at 1528-1531, 2065-2068, 2388-2391, 2771-2775; underfull boxes in Tables 1-2.

The four overfull lines (the worst, 14.4 points, at source lines 2065-2068, inside the compression theorem's statement) should be checked and reflowed, for example by allowing a line break in the displayed infimum or rephrasing. The systematic underfull boxes in the two p-column tables are the standard artifact of justified narrow columns; they are harmless but produce a visibly ragged right edge in the compiled PDF.

Recommendation. Use ragged-right alignment in those table columns (or the `ragged2e` package's moderate variant), and reflow the four overfull lines. Note also that Figure 2 is pushed left with negative horizontal space (line 1105) and Figure 4 is wrapped in a resize box (line 2622); both signal that the figures are slightly too wide for the 2.5 cm margins, and a small width reduction would remove the need for both compensations.

A12 - Bibliography order, unused labels, and an implicit package dependency

Severity: Low. Location: Channel paper, lines 3523-3643; label scan; line 2622.

The manual bibliography is in neither order of first citation nor alphabetical order; for example, the Nielsen-Chuang reference is the first citation in the running text (line 65) but appears fifteenth in the list. Many journals require one convention or the other, and citation-order is trivially achievable with the existing manual thebibliography by reordering entries. Separately, 24 labels are defined but never cross-referenced, including several results that have natural reference points: the qubit worked example is precisely an application of the full-width-interval corollary, and the perfect-separation-indices corollary is the natural citation for the conclusion's open problem 2. Finally, `resizebox` is used at line 2622 while `graphicx` is not explicitly loaded; the compile succeeds through the TikZ/PGF chain, but a directly used command should have its package loaded explicitly.

Recommendation. Reorder the bibliography per the target venue's convention, add cross-references to (or prune) the orphaned labels, and add `usepackage of graphicx` to the preamble.

Additional micro-observations on Manuscript A

Three smaller items were noted and are recorded here for completeness. The Example environment is declared under the remark theorem style (lines 24-26) while sharing the theorem counter, so examples are visually remark-like but numerically theorem-like; either convention is fine, but the current mix is unusual. The introduction's phrase “equivalently a finite quantum comb and quantum strategy” (line 107) reads as equating the comb and strategy formalisms, which are equivalent but distinct frameworks; a phrasing such as “equivalently, a finite quantum comb, or quantum strategy” removes the ambiguity. And the finite-sample theorem's diameter cap (“capped at 2”) appears only inside the proof (line

2984) rather than in the statement; writing the bound as the minimum of two and the displayed quantity in the theorem statement itself would make the statement self-contained.

6. Findings - Manuscript B (Instruments Paper)

Manuscript B is the instruments paper, “Exact Affine Geometry, Optimal Centres, and Certified Compression Bounds for Finite-Outcome Quantum Instruments.” Its mechanical health (compilation, cross-references, citations) is described in Section 3 and is spotless; its quantitative content was verified as recorded in Section 4. The findings below are fewer and milder than for Manuscript A, consistent with the paper's more disciplined notation policy (it contains the explicit notation remark that the channel paper lacks).

B1 - “Kolmogorov n-widths” phrasing collides with the outcome count

Severity: Medium. Location: Instruments paper, line 928 (introduction of the width definition section).

The sentence reads: “The widths defined below are continuous-encoding analogues of the classical Kolmogorov n-widths of a compact set in a normed space, with the linear subspaces of the classical theory replaced by arbitrary continuous images of $\mathbb{R}^r$.” Two issues arise. The symbol n is the paper's outcome count everywhere else, so invoking “n-widths” in the classical sense invites exactly the collision that the paper's own notation remark works to prevent. And the encoders map into $\mathbb{R}^r$; the code spaces are subsets of $\mathbb{R}^r$, not “images of $\mathbb{R}^r$.”

Recommendation. Rephrase as “analogues of the classical Kolmogorov widths, with the approximating linear subspaces replaced by arbitrary continuous encodings into $\mathbb{R}^r$.” The Pinkus reference is exactly right and should stay.

B2 - External citation records to verify

Severity: Medium. Location: Instruments paper, bibliography (lines 1563-1613); full list in Section 8.

Internal citation integrity is perfect (16 keys, 16 entries, all resolving in both directions). Two entries of this paper carry specific verification flags: the Kubicki-Palazuelos-Perez-Garcia entry lists the author as “A. M. Kubicki” and spells the title “no-programing” with a single m , both of which should be checked against the published record (the single- m spelling does occur in the literature, but it should be matched exactly); and the “Realization schemes for quantum instruments” entry's author list and journal data should be confirmed. Section 8 lists these together with the channel paper's items.

Recommendation. Verify each flagged record against the publisher's page before submission. None of them affects the mathematics.

B3 - Possessive-style typography in a table caption

Severity: Low. Location: Instruments paper, line 1454 (caption of the channel-versus-instrument comparison table, Table 2).

The caption states the maximality argument “uses the eigenvalue $1/(n d_A d_B) - t's/(2 d_A) < 0$ for $t' > t$.” The apostrophe-s construction “t’s” is intended as the product of t -prime and s , but it reads as a possessive. The companion proof passage at line 280 uses the same expression without the ambiguity.

Recommendation. Re-typeset as “ $1/(n d_A d_B) - t' s/(2 d_A)$ ” or “ $- s t'/(2 d_A)$ ”.

B4 - Cryptic closing sentence in the tunable-mixing remark

Severity: Low. Location: Instruments paper, lines 736-743 (tunable mixing parameter remark).

The remark correctly explains that the mixing parameter in the full-direction construction can be made direction-dependent, raising the pointwise separation, and honestly notes why the constant value is used in the proof. Its closing sentence, however, states that “a variable quotient $v(Z)$ would require a separate gluing argument”; the correction state $v(Z)$ is in fact well-defined for $d_A \geq 2$, and what actually degenerates is the correction weight $(1 - \lambda(Z))$ tending to zero as the common marginal approaches the maximally mixed state. A reader must reverse-engineer this.

Recommendation. Replace the closing clause with one sentence stating the actual degeneracy: near marginals equal to the maximally mixed state, the correction weight vanishes and continuity of the variable-parameter construction becomes delicate.

B5 - Inline combinatorial argument before a definition

Severity: Low. Location: Instruments paper, lines 1141-1145, immediately before the upper-envelope definition.

The balanced-partition exchange argument (if two block sizes differ by at least two, transferring one dimension strictly decreases the squared sum) is presented as unnumbered prose between the pinching theorem and the upper-envelope definition. It is a genuine lemma with a proof, and its placement as floating prose slightly obscures the structure of Section 5. The channel paper handles the identical argument identically (its lines 1796-1803), so the two papers are at least consistent with each other.

Recommendation. Number it as a lemma (or fold it into the proof of the pinching corollary where it is used). Low priority; consistent across both papers as-is.

B6 - Typesetting warnings in the instruments paper

Severity: Low. Location: Instruments paper, compile log; overfull lines at 49-50, 712-714, 747-748, 1557-1558; underfull boxes in Tables 1-2.

The overfull lines are small except at source lines 712-714 (11.6 points, inside the full-direction theorem's statement); the same reflow advice as finding A11 applies. The underfull boxes concentrate in the two wide p-column tables, and the caption of the comparison table is long enough that the paper might benefit from moving part of it into the body text.

Recommendation. Ragged-right columns for the wide tables; reflow the 712-714 paragraph; consider trimming the comparison-table caption (part of it duplicates finding B3's material and could move to the appendix where the witnesses are constructed).

B7 - Strengths worth preserving

Severity: Informational. Location: Instruments paper, passim.

Several features of this manuscript are exemplary and should survive any revision unchanged: the exact-versus-certified remark (lines 157-160) that itemizes what is proved exactly versus what is only certified; the notation remark (lines 162-165) that de-collides six symbol families before they can collide; the physical-origin remark (lines 303-306) that carefully scopes the $1/n$ factor to the uniform centre and explicitly denies a universal $1/n$ law; the nonconstructive-decoder remark (lines 1061-1064) that states plainly that the Helly-selected decoder need not be continuous or algorithmic; the error-one-barrier remark (lines 1274-1279) that scopes the barrier to the three specific constructions and explicitly does not claim it binds all codes; and the degenerate-case handling (lines 149-155). The instrument-level join proposition (lines 816-841) together with its “why at most one input-dependent sphere” remark (lines 843-846) is a genuinely careful piece of mathematical writing. None of this needs attention; it is recorded here so that revision energy is spent on the defects instead.

B8 - Data Availability spelling

Severity: Low. Location: Instruments paper, line 1561.

The statement uses “analysed” (British), consistent with the rest of this paper, while the channel paper's statement uses “analyzed” (American). This is the smallest instance of the cross-paper convention mismatch recorded as finding A4 and recommendation item on harmonization.

Recommendation. Fold into the A4 harmonization pass.

7. Cross-Paper Consistency Review

The two papers make strong mutual claims: the channel paper's introduction states that the instruments paper extends the ball and compression theorems to finite-outcome instruments and that the channel-level theorems are recovered by taking $n = 1$, while the instruments paper states the converse recovery and points to the channel paper for the general source-latent theory. Every one of these claims was checked against the actual content of the companion, and all of them hold: the affine dimension, the centred and global in-radii, the covering radius and centre-optimality, the replacement-sphere index, the observable-sphere index (with the $n = 1$ case correctly requiring $d_B \geq 2$), the join index with its block-count constant, the full-direction separation $2/d_A$, the intermediate floor as the maximum of the ball and input-dimension terms, and the exact constant-code width all specialize at $n = 1$ to the channel paper's values, line by line. The anonymous companion cross-citations are symmetric (“submitted concurrently, 2026” in both) and every theorem-level pointer (observable sphere, full-direction theorem, join, global in-radius, covering) resolves to a real result with matching content.

One nuance deserves attention. The channel paper's introduction (line 128) says the companion “treats mixed and unequal-output programming” as if it were a main contribution. The instruments paper does contain exactly that material (mixed-program no-programming and the unequal-output corollary), but frames it as a logically independent background appendix that the main theorems do not rely on (line 1493), and its scope remark explicitly points the quantitative programming statements (the positivity floor, program separation, and packing bounds) back to the channel paper. The channel paper's phrasing is not false, but it slightly mislocates the material's weight, and a referee sent looking for programming results among the companion's main theorems will not find them there.

Recommendation. Reword the channel paper's line 128 to something like “records, in an appendix, the no-programming background - including mixed-program and unequal-output cases - and extends the ball and compression theorems to finite-outcome quantum instruments.” Beyond this, the presentation asymmetries catalogued in finding A10 are the remaining cross-paper work: harmonizing Acknowledgements, hyperlink style, definition numbering, the error-one-floor statement, the continuous-decoder threshold refinement, and the analysed/analyzed spelling. A final observation in the opposite direction: the two papers prove the global in-radius optimality by genuinely different routes (halfspace concavity plus Haar twirl in the channel paper; Minkowski-sum concavity plus the enlarged group average in the instruments paper). Both are correct, and the difference is legitimate, but a one-line remark in each paper acknowledging the alternative route would pre-empt the obvious referee question of which proof is intended to be canonical.

8. Citation Verification Notes

Internal citation integrity is perfect in both papers: every key resolves to a bibliography entry and every entry is cited at least once, including the one citation with an optional argument. What this review could not do offline is verify the bibliographic data of the external references against the published records. The items below were flagged during the line-level read, each for a specific reason; all are plausible as cited, and none affects the internal mathematics of either paper. The single exception to “none affects the mathematics” is the first row: the Nechita et al. corollary is load-bearing for Lemma 3.2, and Section 4.3 of this report records an independent re-derivation of that inequality, so the mathematics is covered either way.

Table 4. Bibliographic items to verify before submission

Key (paper)	As cited	What to verify
NechitaEtAl2018 (A)	I. Nechita, Z. Puchala, L. Pawela, K. Zyczkowski, “Almost all quantum channels are equidistant”, J. Math. Phys. 59, 052201 (2018); cited as Corollary 2	That the cited corollary states exactly the upper bound of Lemma 3.2 for Hermiticity-preserving maps in the unnormalized Choi convention; otherwise cite as “see also” and use the self-contained proof of Section 4.3.
BatsonEtAl2021 (A)	J. Batson, C. G. Haaf, Y. Kahn, D. A. Roberts, “Topological obstructions to autoencoding”, J. High Energy Phys. 2021, 280	Article number 280 and the author list.
KvalheimSontag2024 (A)	M. D. Kvalheim, E. D. Sontag, “Why should autoencoders work?”, Trans. Mach. Learn. Res. (2024)	Exact title and venue/year.
WuEtAl2024 (A)	J. Wu, H. Fu, M. Zhu, H. Zhang, W. Xie, X.-Y. Li, “Quantum circuit autoencoder”, Phys. Rev. A 109, 032623 (2024)	Full author list and bibliographic data.
Caro2024 (A)	M. C. Caro, “Learning quantum processes and Hamiltonians via the Pauli transfer matrix”, ACM Trans. Quantum Comput. 5, Article 14 (2024)	Journal, volume, and article number.
SzarekBengtssonZyczkowski2006 (A)	S. J. Szarek, I. Bengtsson, K. Zyczkowski, J. Phys. A 39, L119 (2006)	Author list and title.

Key (paper)	As cited	What to verify
Munkres2000 (A)	J. R. Munkres, Topology, 2nd ed., cited for invariance-of-dimension and invariance-of-domain (line 1216)	That the cited edition actually covers these results; they are more commonly sourced from differential-topology texts - the paper already cites Hirsch, so citing both is a safe fix.
KubickiEtAl2019 (B)	A. M. Kubicki, C. Palazuelos, D. Perez-Garcia, "Resource quantification for the no-programing theorem", Phys. Rev. Lett. 122, 080505 (2019)	Author initials, and the single-m spelling "programing" against the published title.
ChiribellaEtAl2009 (B)	G. Chiribella, G. M. D'Ariano, P. Perinotti, "Realization schemes for quantum instruments", J. Math. Phys. 50, 042101 (2009)	Author list and exact title/journal data for this specific paper.

The two companion cross-citations are internally consistent by construction and need no verification beyond what the authors control. The remaining references (Choi, Jamiofkowski, Watrous, Matousek, Barvinok, Munkres as a book, Hirsch, DeVore-Howard-Micchelli, Cohen et al., Petrova-Wojtaszczyk, Nielsen-Chuang, Yang-Renner-Chiribella, Chiribella et al. 2009 (networks), Gutoski-Watrous, Scott, Haah et al., Davies-Lewis, Ozawa, Aharonov-Kitaev-Nisan, Helstrom, and Pinkus) are standard works whose data as cited matched this reviewer's knowledge to the level of detail that can be checked offline.

9. Prioritized Recommendations

The table below consolidates all recommendations from Sections 5 through 8 into a single priority-ordered list. Priorities reflect the likely referee and editor impact: priority 1 items are internal inconsistencies visible within minutes of reading; priority 2 items are verification and wording matters that affect trust in load-bearing steps; priority 3 items are editorial polish and companion harmonization; priority 4 items are optional improvements. All fixes are editorial in the strict sense: no theorem needs re-derivation, no proof needs its mathematics changed, and no result is weakened by any recommendation.

Table 5. Consolidated recommendations

Priority	Action	Where
P1	Redraw or relabel the flag-space node in Figure 2 (remove C_n/n or define them)	A, lines 1104-1147
P1	Unify spelling within the channel paper and align the two papers' conventions	A passim; both Data Availability statements

Priority	Action	Where
P1	Disambiguate s, q, and Delta; add a notation remark mirroring the companion's	A, lines 782/3380/1143, 1161/3380, 472/746/2585/2887
P2	Add the self-contained proof of Lemma 3.2's upper bound; verify the Nechita et al. corollary	A, lines 712-734
P2	Verify the flagged bibliographic records (Table 4)	Both papers
P2	Reword "Matching upper bounds" in the abstract	A, line 50
P3	Fix the duplicated decoder sentence; split the compound Clifford theorem; re-typeset "t's" in the comparison-table caption; clarify the tunable-mixing remark; fix the Kolmogorov-width phrasing	A lines 1762-1764, 1159-1169; B lines 1454, 736-743, 928
P3	Harmonize companion presentation: Acknowledgements, hyperref style, definition numbering, an error-one-floor corollary, the continuous-decoder threshold statement, table rule style	Both papers
P3	Silence table underfull warnings (ragged columns) and reflow the overfull lines in each paper	A lines 1528-1531 etc.; B lines 712-714 etc.
P4	Demote or justify the structured-partition appendix; reference or prune the 24 unused labels; state the $\min\{2, \cdot\}$ cap in the finite-sample theorem; load graphicx explicitly; reorder the bibliography per venue style	A, various

Sequencing advice: the three priority-1 items and the first two priority-2 items can all be completed in a single focused editing session, and they remove every issue that a fast-reading referee could weaponize. The priority-2 citation checks are independent of the editing work and can proceed in parallel. Priority-3 harmonization is best done once, deliberately, after the two manuscripts are otherwise final, so that the pair is consistent at submission time.

10. Concluding Assessment

Both manuscripts are, on the evidence of this line-level review, submission-ready in their mathematics. Every constant recomputed matched its source; every proof re-derived held;

the compile and integrity checks are clean in both papers; and the honest separation of exact results from certified bounds is maintained consistently enough that a referee can trust the paper's own account of what is proved. The open problems each paper declares (the intermediate positive widths, the perfect-separation indices, the uniqueness of the optimal centre, and the boundary cases of the scaling trichotomy) are stated as open honestly, and nothing in this review suggests any of them is quietly claimed.

The work that remains is presentation discipline, and it is modest. The channel paper needs one figure re-labeled, its notation de-overloaded, and its spelling unified; both papers need their bibliographic records verified and roughly a dozen small editorial fixes; and the pair needs one harmonization pass so that they look like the coordinated co-submissions they are. Completing the priority-1 and priority-2 items in Section 9 would leave both manuscripts free of any issue this review could find at line level, with the exception of the deliberately open mathematics, which is exactly where a paper should be open.